

A comparison of two physics-based numerical models for simulating surface water–groundwater interactions

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ABSTRACT

Problems in hydrology and water management that involve both surface water and groundwater are best addressed with simulation models that can represent the interactions between these two flow regimes. In the current generation of coupled models, a variety of approaches is used to resolve surface–subsurface interactions and other key processes such as surface flow propagation. In this study we compare two physics-based numerical models that use a 3D Richards equation representation of subsurface flow. In one model, surface flow is represented by a fully 2D kinematic approximation to the Saint-Venant equations with a sheet flow conceptualization. In the second model, surface routing is performed via a quasi-2D diffusive formulation and surface runoff follows a rill flow conceptualization. The coupling between the land surface and the subsurface is handled via an explicit exchange term resolved by continuity principles in the first model (a fully-coupled approach) and by special treatment of atmospheric boundary conditions in the second (a sequential approach). Despite the significant differences in formulation between the two models, we found them to be in good agreement for the simulation experiments conducted. In these numerical tests, on a sloping plane and a tilted V-catchment, we examined saturation excess and infiltration excess runoff production under homogeneous and heterogeneous conditions, the dynamics of the return flow process, the differences in hydrologic response under rill flow and sheet flow parameterizations, and the effects of factors such as grid discretization, time step size, and slope angle. Low sensitivity to vertical discretization and time step size was found for the two models under saturation excess and homogeneous conditions. Larger sensitivity and differences in response were observed under infiltration excess and heterogeneous conditions, due to the different coupling approaches and spatial discretization schemes used in the two models. For these cases, the sensitivity to vertical and temporal resolution was greatest for processes such as infiltration and ponding, although the differences between the hydrographs of the two models decreased as mesh and step size were progressively refined. In return flow behavior, the models are in general agreement, with the largest discrepancies, during the recession phase, attributable to the different parameterizations of diffusion in the surface water propagation schemes. Our results also show that under equivalent parameterizations, the rill and sheet flow conceptualizations used in the two models produce very similar responses in terms of hydrograph shape and flow depth distribution.

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1. Introduction

Surface and subsurface waters are not isolated components of the hydrologic cycle, but instead interact in response to topographic, soil, geologic, and climatic factors [8]. The study of these interactions has been addressed at both small (field and hillslope)

(e.g. [1,39]) and large (watershed to global) scales (e.g. [23,37]). A number of hydrological models that incorporate some representation of groundwater–surface water interactions have been developed over the past decades, including physically-based, distributed-parameter models. This latter class of models, more rigorous but also more computationally intensive than empirical or semi-empirical approaches, uses the shallow water equations to describe surface flow, i.e., one- or two-dimensional approximations of the Saint-Venant equations for overland and/or channel flow, coupled with a subsurface component that solves the

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three-dimensional equation for variably saturated flow, i.e., Richards' equation (e.g. [41,28,33,18]). A comprehensive description of the types of process representation in distributed models and their inherent assumptions and limitations, together with a discussion of comparison and assessment issues, is provided in Kampf and Burges [19], Clarke [5], Furman [10], Ebel et al. [9], and Maxwell [24].

For physically-based coupled models, which are the focus of this study, various schemes have been proposed for solving the system of surface and subsurface equations and for resolving the interactions across the land surface. The solution approaches can be broadly classified as full coupling, sequential coupling, and loose coupling, whereas the formulations for the exchange fluxes are based on continuity principles, diffusion paradigms, boundary conditions switching, or other schemes. In full coupling (e.g. [41,33,20]), the governing equations are solved simultaneously; in sequential coupling (e.g. [12,28,3]), they are solved separately, with an explicit discretization used for at least one of the equations or with an iterative cycle superposed on the overall system; in loose coupling (e.g. [36,6]), the equations are again solved separately, with the output from one regime (e.g., surface flow) simply passed as input to the other, without iteration or other conditions imposed.

Whereas the accuracy, robustness, and other performance features of surface and subsurface numerical models have been extensively documented (e.g. [35,42] for Saint-Venant approximations; [16,29] for Richards' equation), there have been very few assessments of coupled models based on these equations. The purpose of this study is to provide such an assessment via a comparative analysis of two process-based groundwater-surface water models. One model, ParFlow [20,21], uses a full coupling approach and continuity of pressures and fluxes across the land surface to resolve the surface-subsurface interactions; the other model, CATHY [2,3], is based on sequential coupling with boundary condition switching to partition atmospheric fluxes into infiltration (or exfiltration) and a change in surface water storage. A comparison of these two very different models provides a first opportunity to critically examine some key features of coupled hydrological models. In addition to different schemes for coupling and exchange flux resolution, the two models use different conceptualizations of surface routing: sheet flow representation and a kinematic wave equation in ParFlow; rill flow representation and a diffusion wave equation in CATHY. Although not directly inherent to coupling issues, these additional differences are also worthy of assessment, given the high interest in applying coupled hydrological models at catchment and river basin scales where terrain features (slope, roughness, etc.), and consequently surface flow conditions, can vary greatly. Other differences between the models (e.g., ParFlow uses a finite difference/finite volume discretization whereas CATHY uses finite elements for the subsurface and finite differences for the surface) will also have an effect on the intercomparison tests and will be duly considered.

The intercomparison study is carried out through a series of simple test cases subjected to step functions of rainfall followed by a recession or evaporation period. The test cases involve a sloping plane [11] and a tilted V-catchment [33]. The simulations are designed to clearly expose model differences and similarities under complex and realistic physical conditions. The first tests focus on the different treatments of the exchange fluxes between the subsurface and surface domains and their sensitivity to factors such vertical mesh resolution, time step size, and slope angle. A second set of tests is intended to evaluate the impact of the different conceptualizations for propagation of surface runoff in terms of water depth distribution at the ground surface and timing and shape of the hydrograph.

2. Description of the models

The governing equations for the ParFlow model [20,25] are the three-dimensional (3D) Richards equation for subsurface flow in variably saturated soils and the kinematic wave approximation of the Saint-Venant equation for overland and channel flow:

$$S_s S_w \frac{\partial \psi}{\partial t} + \phi \frac{\partial S_w}{\partial t} = -\nabla \cdot \mathbf{q} + q_s, \quad (1)$$

$$\mathbf{q} = -K_s K_r \nabla (\psi - z), \quad (2)$$

$$\frac{\partial \psi_s}{\partial t} = \nabla \cdot (\psi_s \vec{v}) + q_r(x), \quad (3)$$

$$S_{f,i} = S_{o,i}, \quad (4)$$

where S_s is the specific storage coefficient [1/L], $S_w = S_w(\psi)$ is the relative saturation [–], ψ is the subsurface pressure head [L], t is time [T], ϕ is the porosity [–], ∇ is the gradient operator, q is the Darcy flux [L/T], q_s is a general source/sink term [1/T], K_s is the saturated hydraulic conductivity tensor [L/T], $K_r = K_r(\psi)$ is the relative hydraulic conductivity function [–], z is the vertical coordinate pointing downward [L], ψ_s is the pressure at the ground surface (surface ponding depth) [L], $\vec{v}$ is the depth-averaged velocity vector [L/T], q_r is the rainfall rate [L/T], and $S_{f,i}$ and $S_{o,i}$ are the gravity forcing and friction slope terms, respectively [–], with i indicating the x and y directions.

The governing equations for the CATHY model [2,3] are the 3D Richards equation and the diffusion wave approximation of the Saint-Venant equation:

$$S_s S_w \frac{\partial \psi}{\partial t} + \phi \frac{\partial S_w}{\partial t} = \nabla \cdot [K_s K_r(\psi)(\nabla \psi + \eta_z)] + q_{ss}, \quad (5)$$

$$\frac{\partial Q}{\partial t} + c_k \frac{\partial Q}{\partial s} = D_h \frac{\partial^2 Q}{\partial s^2} + c_k q_s, \quad (6)$$

where $\eta_z = (0, 0, 1)^T$ (the vertical coordinate is positive upward), q_{ss} represents distributed source (positive) or sink (negative) terms [L³/L³T], Q is the discharge along the channel link [L³/T], c_k is the kinematic wave celerity [L/T], s is the hillslope/channel link coordinate [L], D_h is the hydraulic diffusivity [L²/T], and q_s is the inflow (positive) or outflow (negative) from the subsurface to the surface [L³/LT].

2.1. Subsurface flow

The three-dimensional Richards equation is solved in ParFlow using a cell-centered finite difference scheme with harmonic averages of the saturated hydraulic conductivity and a one-point upstream weighting of the relative permeability function. CATHY uses a Galerkin finite element spatial integrator using tetrahedral elements and linear basis function. Both models employ an implicit backward Euler scheme for the discretization in time of Richards' equation. The resulting discrete equation is solved in ParFlow by a Newton–Krylov nonlinear solver [17] and in CATHY by Picard or Newton iteration [34].

2.2. Surface routing

The kinematic wave Eqs. (3) and (4) in ParFlow are discretized in space with an upwind finite volume scheme and in time with an implicit backward Euler method. The surface flow equation is solved by posing two types of boundary conditions: the gradient and critical depth outlet conditions, consisting respectively in a prescribed flux and water depth condition. In CATHY the diffusion wave Eq. (6) is discretized in space and time with a matched artificial dispersivity (MAD) scheme [38].

ParFlow uses a two-dimensional sheet flow conceptualization for surface routing, whereby flow is assumed to be distributed in

a uniform sheet over a plane surface. The different dynamics of overland and channel flow are handled by assigning different Manning's roughness coefficients to each regime, and there is no need for parameters related to flow geometry. In CATHY surface routing is conceptualized as rill flow along a one-dimensional drainage network of connected linear segments. This network is automatically extracted by a digital elevation model (DEM)-based pre-processor [31]. To differentiate between overland and channel flow regimes, different hydraulic parameters (resistance/conductance coefficient and water-surface width) and dynamic scaling properties are assigned to hillslope and channel cells, which have been distinguished in the pre-processing step based on threshold criteria linked, for example, to upstream drainage area [26]. Hydraulic characterization of the drainage network is achieved by applying the Leopold and Maddock [22] power law relations that relate water-surface width and roughness coefficient to discharge [30]:

$$W(A, Q) = W(A_s, Q_f) Q_f (A_s)^{-b'} (A/A_s)^{w(b''-b')} \cdot Q^{b'}, \quad (7)$$

$$k_s(A, Q) = k_s(A_s, Q_f) Q_f (A_s)^{-y'} (A/A_s)^{w(y''-y')} \cdot Q^{y'}, \quad (8)$$

where W [L] is the water-surface width, k_s [L^{1/3}/T] is the Gauckler–Strickler coefficient, A_s is the drainage area at a selected site, the exponents b' , b'' , y' , y'' are characteristics of the rill/channel network as a whole, and w [–] is the exponent of the scaling relationship of the selected frequency discharge Q_f with drainage area A . The width and roughness coefficients are thus dynamically varied in space and time, with water-surface width for instance ranging from a present minimum value up to the cell size.

The temporal solution is globally implicit in ParFlow, using backward Euler for both the surface and subsurface equations. The time step size can be constant or it can vary based on an exponential growth method. In CATHY the time step size for the surface routing module is computed based on a Courant number accuracy criterion. The explicit in time nature of the MAD scheme allows sequential resolution of the surface and subsurface equations and a nested time stepping approach, whereby for each subsurface time step, one or more sub-steps (depending on the Courant estimate) are taken for the surface routing module.

2.3. Coupling and resolution of exchange fluxes

As detailed in Kollet and Maxwell [20], ParFlow simulates fully-coupled surface and subsurface flow via an overland flow boundary condition. The approach takes Eq. (3) and adds a flux for subsurface exchanges, $q_e(x)$:

$$\frac{\partial \psi_s}{\partial t} = \nabla \cdot (\psi_s \vec{v}) + q_r(x) + q_e(x). \quad (9)$$

Continuity of pressure at the top cell of the boundary between the surface and subsurface systems is then assigned by setting pressure head ψ in Eq. (1) equal to the vertically-averaged surface pressure ψ_s as follows:

$$p = \psi_s = \psi. \quad (10)$$

This relationship is then substituted back into Eq. (9) as follows:

$$\frac{\partial \|p, 0\|}{\partial t} = \nabla \cdot (\|p, 0\| \vec{v}) + q_r(x) + q_e(x), \quad (11)$$

where the $\|p, 0\|$ operator chooses the greater of the two quantities, p and 0. By posing now the continuity of the fluxes ($q_{bc} = q_e$, where q_{bc} is a boundary condition of Neumann type for the subsurface equation) at the ground surface:

$$-K_s K_r(p) \nabla(p - z) = \frac{\partial \|p, 0\|}{\partial t} - \nabla \cdot (\|p, 0\| \vec{v}) - q_r(x). \quad (12)$$

Eqs. (1) and (2) can be re-written in the following combined form:

$$S_{sw} \frac{\partial \psi}{\partial t} + \phi \frac{\partial S_w(\psi)}{\partial t} = -\nabla \cdot [-K_s K_r(p) \nabla(p - z)] + q_s. \quad (13)$$

When the pressure at the top cell of the subsurface domain has a ponded depth and is thus greater than zero (Dirichlet boundary condition), the kinematic wave equation is active through the left term in Eq. (12) and the depth-averaged velocity calculated through Manning's depth-discharge relationship. On the other hand, when the pressure at the top cell of the subsurface is lower than zero (Neumann boundary condition), the overland flow boundary condition is turned off and Eq. (13) is solved as a standard Richards equation. This approach results in both systems (surface and subsurface) solved over a common grid in a fully-integrated, fully-mass conservative manner.

In CATHY the source/sink terms q_{ss} and q_s in Eqs. (5) and (6) represent the exchanges between the subsurface and surface modules. At any given time t , atmospheric forcing inputs are imposed as land surface boundary conditions on the subsurface equation, which is solved first. After solution, a balance is performed at each surface node between the atmospheric water supply (rainfall) or demand (potential evaporation), the capacity of the soil to accept (infiltrate) or deliver (exfiltrate) this water, and the amount of water already present at the surface (ponding). The boundary condition is updated to reflect whether, after this balance, the soil capacity exceeds or not the water supply (or demand), including any water held as surface storage (ponded). Exceedance implies a Neumann (specified flux) boundary condition, equivalent to an atmosphere-controlled process, whereas in the converse case a Dirichlet (specified head) condition applies and the process is soil-limited. This boundary condition switching procedure is an extension to the case of ponding (and thus interaction with a surfacing routing model) of the method commonly used in Richards' equation-based subsurface models to perform rainfall-runoff partitioning and to restrict water loss when the soil becomes too dry. One of the appealing features of this procedure is that it automatically tracks both infiltration excess (Hortonian) and saturation excess (Dunnian) runoff generation mechanisms. In the ponding extension, the parameter *Pond_head_min* is introduced in CATHY as the threshold pressure head (or ponding) value a surface node must attain before water can be routed by the surface flow module. This parameter represents, for instance, the amount of water that can be stored in microtopographic depressions on the land surface.

To illustrate the boundary condition switching approach consider a rainfall event on an initially unsaturated soil. The surface boundary condition will be of Neumann type until the pressure head at the surface, computed by the model, becomes zero. At this time the boundary condition is switched to Dirichlet type, allowing the model to compute the actual rate at which water enters the soil. If during the course of the rainfall period the actual flux becomes larger than the potential rate (as could happen for instance if there is a fall in precipitation intensity), the boundary condition is switched back to Neumann type. In the interaction with the surface module, any water that accumulates at the surface (in excess of *Pond_head_min*) when the boundary condition is of Dirichlet type becomes available for routing.

Switching between Neumann and Dirichlet boundary conditions according to the saturation status of a surface node, as done in CATHY, resembles the way in which the boundary condition Eq. (12) in ParFlow takes on different forms depending on whether the surface pressure head is positive or not. A key difference in the coupling approach for the two models is that, by embedding the surface flow equation into the subsurface equation, ParFlow resolves the exchange between surface and subsurface at the same time level, whereas in CATHY, where the explicit nature of the MAD scheme allows sequential and noniterative resolution of the coupled system, there is a time level lag when solving the surface equation following the subsurface one.

3. Test case descriptions

In the sloping plane simulations, surface–subsurface interactions were investigated for the infiltration excess (Horton) and saturation excess (Dunne) runoff generation mechanisms under homogeneous and heterogeneous conditions, and for the return flow process under homogeneous conditions. Rill flow vs. sheet flow routing for overland flow was also examined for this test case (channel flow was not considered for this test). The test catchment (Fig. 1) is 400 m long by 320 m wide. A surface grid of 5×4 cells, or 80 m resolution, was used for the homogeneous tests of Horton and Dunne processes and for the overland flow cases, a grid of 40×32 , or 10 m resolution, was used for the heterogeneous tests, and a grid of 80×64 , or 5 m resolution, was used for the return flow cases. The soil depth is 5 m and van Genuchten relationships [40] are used to describe the soil hydraulic characteristics, with saturated moisture content $\theta_{sat} = 0.4$, residual moisture content $\theta_{res} = 0.08$, $\alpha = 1.0$, and $n = 2.0$. For CATHY the Gauckler–Strickler conductance coefficient (k_s) was kept fixed in space and time at $3.0 \times 10^3 \text{ m}^{1/3}/\text{min}$ (i.e., $y' = y'' = 0$ in Eq. (8)) and the exponent w was kept fixed at 1.0. This parameterization corresponds in ParFlow to a Manning's roughness coefficient of $3.3 \times 10^{-4} \text{ m}^{-1/3} \text{ min}$. In order to have a consistent basis for intercomparison, the coupled simulations in the runoff generation tests were run with the CATHY model parameterized as sheet flow, using a constant water-surface width (Eq. (7)) equal to the grid resolutions for the 80 m and 10 m grids. The bottom and vertical sides of the domain are no-flow boundaries for the subsurface in both models. ParFlow uses a gradient condition at the catchment outlet for the overland flow simulator, while CATHY, with an overland simulator based on a hyperbolic equa-

tion, does not need to specify a boundary condition at the outlet. The simulation consists of a 200-min rainfall event (at a uniform intensity of $3.3 \times 10^{-4} \text{ m/min}$) followed by 100 min of drainage for the runoff generation and overland flow tests, whereas for the return flow cases a 200 min rainfall event (at a uniform intensity of $1.5 \times 10^{-4} \text{ m/min}$) followed by 200 min of evaporation (at a uniform rate of $5.4 \times 10^{-6} \text{ m/min}$) was used. For the runoff generation tests two inclinations of the plane (in the x direction) are used, of 0.05% and 5% (the slope in the y direction is zero), together with a range of uniform vertical discretizations from 0.0125 m (400 layers) to 0.2 m (25 layers). For the return flow cases inclinations of 0.5% and 5% and a uniform vertical discretization of 0.05 m (100 layers) are used, and for the overland flow tests the slope inclination is 0.05%. Reported outflow rates for the two models are with reference to the grey strip of cells shown in Fig. 1.

In the tilted V-catchment only surface flow processes are considered, but we now consider both overland and channel flow. The catchment consists of two identical sloping planes of length 800 m and width 1000 m joined by a 20 m wide, 1000 m long channel. The x direction follows the hillslope length and channel width while the y direction is along the channel length and hillslope width (Fig. 1). The Gauckler–Strickler conductance coefficient is constant in space and time at $4.0 \times 10^3 \text{ m}^{1/3}/\text{min}$ for hillslope cells and $4.0 \times 10^2 \text{ m}^{1/3}/\text{min}$ for channel cells, which correspond in ParFlow to a Manning's roughness coefficient of $2.5 \times 10^{-4} \text{ m}^{-1/3} \text{ min}$ and $2.5 \times 10^{-3} \text{ m}^{-1/3} \text{ min}$, respectively. The simulation consists of a 90-min rainfall event (at a uniform intensity of $1.8 \times 10^{-4} \text{ m/min}$) followed by 90 min of drainage.

A summary of the parameter values for the different test cases is given in Tables 1 and 2.

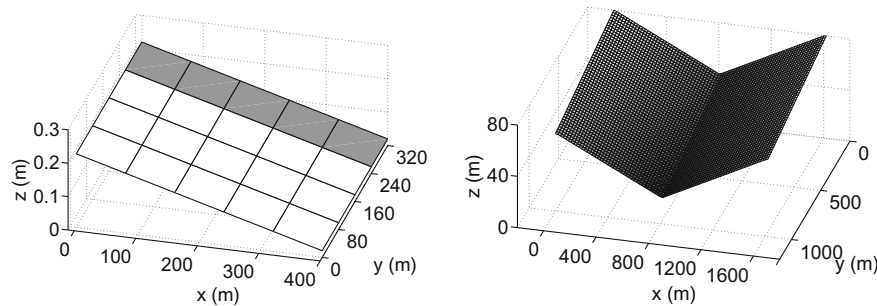


Fig. 1. The sloping plane catchment (left) and the tilted V-catchment (right). Model outflow results for the sloping plane are for the grey strip of cells.

Table 1

Parameter values for the sloping plane and tilted V-catchment test cases.

	Sloping plane				V-catchment
	Homogeneous			Heterogeneous	
	Saturation excess	Infiltration excess	Return flow	Slab	
Horizontal mesh size, $\Delta x = \Delta y$ (m)	80	80	5	10, 1	20
Vertical mesh size, Δz (m)	0.0125, 0.025, 0.05, 0.1, 0.2	0.0125, 0.025, 0.05, 0.1, 0.2	0.05	0.05	na
Water table depth at time 0, WT (m)	0.5, 1.0	1.0	0.5	1.0	na
Saturated hydraulic conductivity, K_{sat} (m/min)	6.94×10^{-4}	6.94×10^{-6} , 6.94×10^{-5}	6.94×10^{-2}	6.94×10^{-4} , 6.94×10^{-6}	na
Rainfall rate, q_r (m/min)	3.3×10^{-4}	3.3×10^{-4}	1.5×10^{-4}	3.3×10^{-4}	1.8×10^{-4}
Evaporation rate, q_e (m/min)	0	0	5.4×10^{-6}	0	0
Gauckler–Strickler conductance coefficient, k_s ($\text{m}^{1/3}/\text{min}$):					
– Hillslope cells	3.0×10^3	3.0×10^3	3.0×10^3	3.0×10^3	4.0×10^3
– Channel cells	na	na	na	na	4.0×10^2
Manning's roughness coefficient, n ($\text{m}^{-1/3} \text{ min}$):					
– Hillslope cells	3.3×10^{-4}	3.3×10^{-4}	3.3×10^{-4}	3.3×10^{-4}	2.5×10^{-4}
– Channel cells	na	na	na	na	2.5×10^{-3}

Table 2
CATHY surface parameters for the tilted V-catchment test case.

	$W(A_s, Q_f)$ [m]	A_s [m ²]	Q_f [m ³ /min]	b', b'' [l]
Sheet flow:				
– Hillslope and channel cells	20	1.62×10^6	291.6	0, 0
Rill flow:				
– Hillslope cells	6.5	1.6×10^4	2.88	0.36, 0.50
– Channel cells	20	1.62×10^6	291.6	0.36, 0.50
Combined rill/sheet flow:				
– Hillslope cells	20	1.6×10^4	2.88	0.36, 0.50
– Channel cells	20	1.62×10^6	291.6	0, 0

4. Results and discussion

4.1. Sloping plane

4.1.1. Runoff generation mechanisms, homogeneous conditions

Surface–subsurface water exchanges under Dunne and Horton saturation processes were first investigated for the homogeneous case. In the saturation excess test two initial water table configurations, at 0.5 m and 1.0 m from the ground surface, were simulated with a saturated hydraulic conductivity K_s of 6.94×10^{-4} m/min. In the infiltration excess test the initial water table depth was fixed at 1.0 m and two K_s values were used, 6.94×10^{-5} and 6.94×10^{-6} m/min. The influence of vertical mesh size resolution is explored for both processes by varying the vertical discretization from 0.0125 m (400 layers) to 0.2 m (25 layers).

Fig. 2 shows the ParFlow and CATHY outflow rates for the two sub-cases in each of the direct runoff tests. In the saturation excess case the results for the two models compare quite well, with an excellent match in the timing and shape of the recession portion of the hydrograph and with percentage differences in total outflow volume (relative to ParFlow totals) ranging from 1.4% (finest discretization, 0.0125 m, and 0.5 m initial water table depth) to 9% (coarsest discretization, 0.2 m, and 1 m initial water table depth). The CATHY results show a slight lag with respect to ParFlow in the rising portion, becoming more pronounced for the deeper initial water table configuration. This discrepancy is due to the slight difference in time to ponding. There is little sensitivity to vertical discretization in both models for the saturation excess case. In the infiltration excess case ParFlow and CATHY again agree quite well, especially in the time to peak and in the recession curve. Total outflow differences for this case range from 2.7% (higher conductivity, 0.0125 m discretization) to 23% (lower conductivity, 0.2 m discretization). There is a greater sensitivity to vertical grid resolu-

tion for both models in this case, and in particular for ParFlow, illustrating the importance of a fine near-surface mesh in accurately capturing the advent (rising limb) of Horton runoff generation. The reasons for this sensitivity to Δz are however different for the two models. In ParFlow (finite difference discretization), the solutions differ because of the finite storage volume that must be saturated at the top model layer for ponding to occur, resulting in a shift in time in the rising limbs of the hydrographs. In CATHY (finite element discretization), a thicker top layer element (larger Δz) will be drier at its base than a thin element during an infiltration excess process, resulting in a lower hydraulic conductivity (since K_r is calculated from the nodal values over that element), resulting in a smaller volume of infiltrated water.

In Fig. 3 it is verified that the numerical results converge as the vertical resolution is refined. Plotted in this figure are relative differences in outflow as a function of resolution expressed as total discharge volume. These are computed with respect to the solution obtained at the finest mesh ($\Delta z = 0.0125$ m) as:

$$\varepsilon_i = \sqrt{\sum_{j=1}^N [(Q_j^{(i)} - Q_j^{(f)}) \Delta t]^2}, \quad (14)$$

where $Q^{(i)}$ is the flow discharge for the i th vertical discretization, $Q^{(f)}$ is the flow discharge for the finest discretization, N is the number of time steps, and Δt is the step size (the same constant Δt was used for all runs). Note that the true solution is unknown. We show only the differences between solutions obtained at a given resolution with that of the finest resolution simulated by that code. In this way, we may use ε as a quantitative measure to test the relative effect of vertical resolution. The results show that the differences are lower for the saturation excess mechanism and larger for the smaller conductivity sub-case (bottom-left graph in Fig. 3), i.e., the near-surface vertical discretization needs to be finer as the infiltration excess mechanism becomes more prominent. Vertical discretization effects are also explored in Downer and Ogden [7], for a Richards equation subsurface model, and related issues connected to coupling formulations are addressed in Ebel et al. [9].

In Fig. 4 the influence of temporal discretization is examined for the 0.05% slope angle of the previous tests with the finest vertical discretization (i.e. $\Delta z = 0.0125$ m), and also for a much steeper sloping plane (5%). Very good agreement between the ParFlow and CATHY models is obtained for the entire time step size range and for both Horton and Dunne surface–subsurface exchange processes.

For ParFlow, a constant time step size of 1, 5, and 10 min was used during the whole simulation for both runoff processes. For

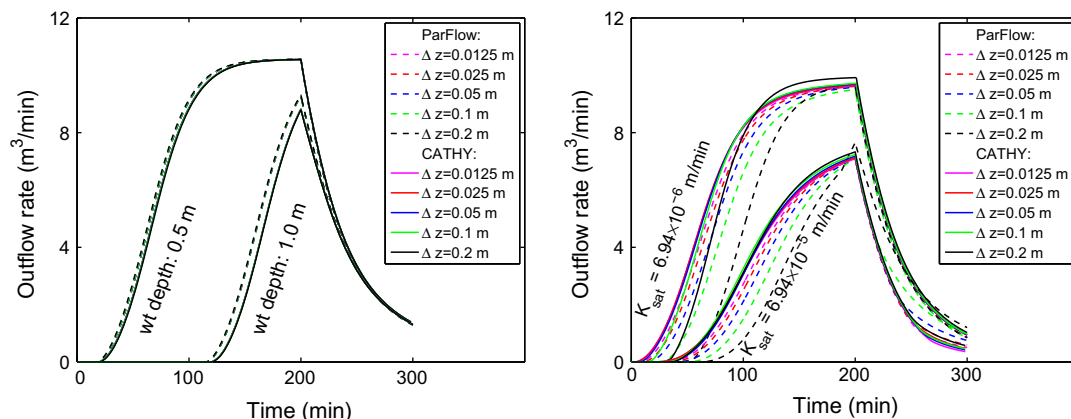


Fig. 2. Outflow response for the homogeneous sloping plane test case under conditions of saturation excess runoff generation with two initial water table configurations (left) and under conditions of infiltration excess runoff generation with two values of saturated hydraulic conductivity (right).

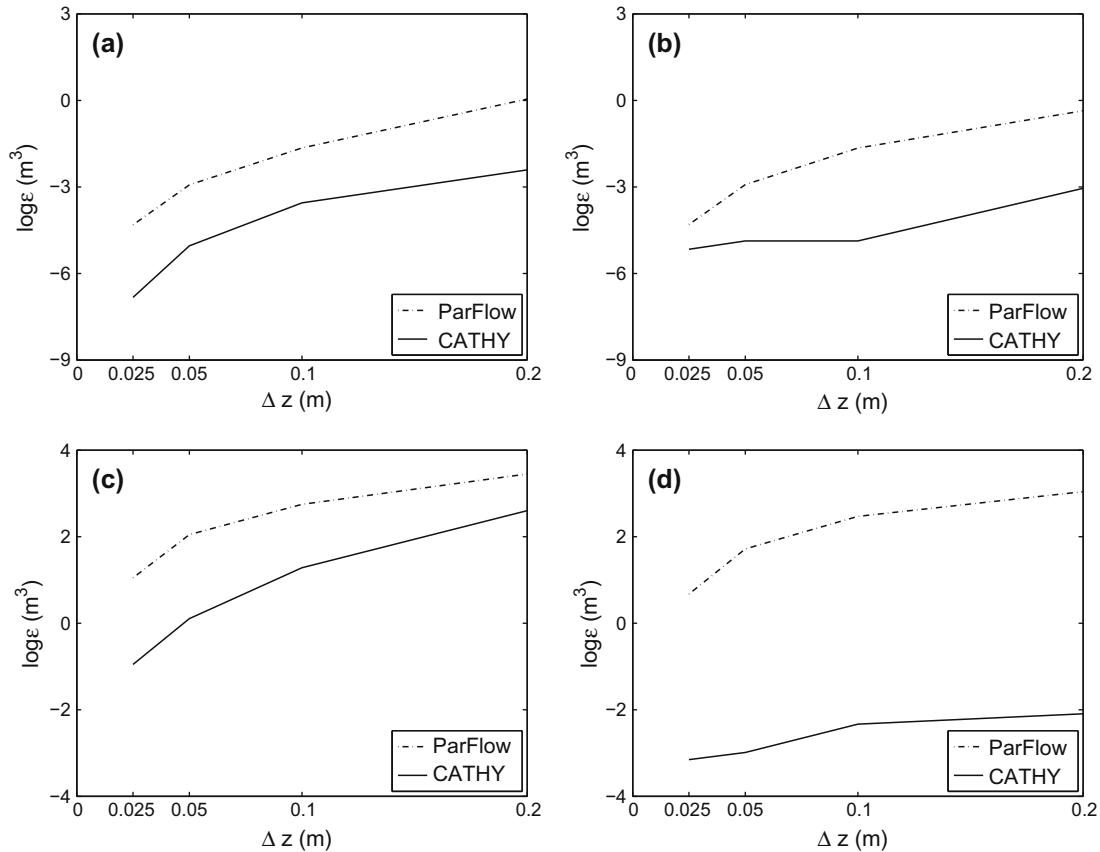


Fig. 3. Relative discharge differences, as defined in Eq. (14), as a function of vertical discretization for the homogeneous sloping plane test case. Top: saturation excess runoff generation with initial water table depth of 0.5 m (a) and 1 m (b). Bottom: infiltration excess runoff generation with saturated hydraulic conductivity $K_s = 6.94 \times 10^{-6}$ m/min (c) and $K_s = 6.94 \times 10^{-5}$ m/min (d).

CATHY, a constant step size was used for the $\Delta t = 1$ min case; for the other simulations, given an initial, minimum, and maximum (Δt_{max}) time step size, the step size for the subsurface module was adapted based on the convergence behavior of the iterative scheme (Picard in this paper) that is used to linearize Richards' equation [34]. When convergence is rapid (less than five iterations in this paper) the step size is increased, and when it is slow (more than 10 iterations) Δt is decreased. A higher Δt_{max} in CATHY generally results in a higher average subsurface time step size (Δt_{avg}) for the overall simulation, but at the possible expense of a greater number of nested time steps to solve the surface routing equation. For the results reported in Fig. 4 this tradeoff was only apparent for the saturation excess case, and somewhat significant only for the higher slope angle (as could be expected, since the faster surface routing in this case requires smaller time steps). The maximum number of nested time steps for each of the CATHY runs in Fig. 4 is as follows: one (i.e., no nesting needed) for the smallest Δt_{avg} at both slope angles for both the Horton and Dunne cases; two for the higher Δt_{avg} for both the Horton and Dunne case at the lower slope; seven for the highest Δt_{avg} for the Horton case at the higher slope angle; eight for the highest Δt_{avg} for the Dunne case at the higher slope angle.

4.1.2. Runoff generation mechanisms, heterogeneous conditions

For the heterogeneous sloping plane the slope angle was 0.05%, the vertical discretization was 0.05 m (100 layers), and the initial water table depth was 1 m. A 100 m long slab of low conductivity ($K_s = 6.94 \times 10^{-6}$ m/min) was placed in the middle of the slope (from $x = 150$ m to $x = 250$ m, for all y). The slab was 0.05 m thick and occupied the first layer. The hydraulic conductivity outside

of the slab was 6.94×10^{-4} m/min. Two different DEM resolutions were used, 10 m and 1 m, and simulations were conducted using a constant time step size (1.0 min and 3.0 min) for ParFlow and variable (adaptive) time stepping ($\Delta t_{avg} = 1.0$ min and 3.0 min) for CATHY. All other parameters and simulation conditions are as reported for the homogeneous sloping plane.

The stepped hydrograph shown in Fig. 5 is a result of the different infiltration–runoff dynamics occurring upslope and downslope of the slab and on the slab itself. Reinfiltration of water that runs off the slab causes earlier saturation of the downslope portion, producing the first plateau in the hydrograph, while later saturation of the upslope portion contributes to the second peak. The reasonable similarity of the curves plotted in Fig. 5 demonstrates that both models are able to capture surface–subsurface interactions for this difficult test problem, while at the same time the discrepancies highlight the algorithmic differences between the coupling approaches used in ParFlow and CATHY. Simultaneous solution of the surface and subsurface equations in ParFlow produces a sharper, more localized response that is less influenced by the temporal discretization. In CATHY, the sequential solution procedure implies, for instance, that once the slab surface becomes saturated (and the boundary condition for these nodes is switched from Neumann to Dirichlet), all of this water is routed downslope for the duration of one time step, delaying any possible reinfiltration until the next (subsurface) time step. Thus the CATHY hydrograph appears a little bit earlier than ParFlow's, and the outflow for the first rising limb is more disperse. A similar situation is observed in the subsurface, as shown in Fig. 6. The greater spreading of the reinfiltration front for CATHY early in the simulation is apparent, and indeed at the coarser DEM resolution this spreading can also be seen

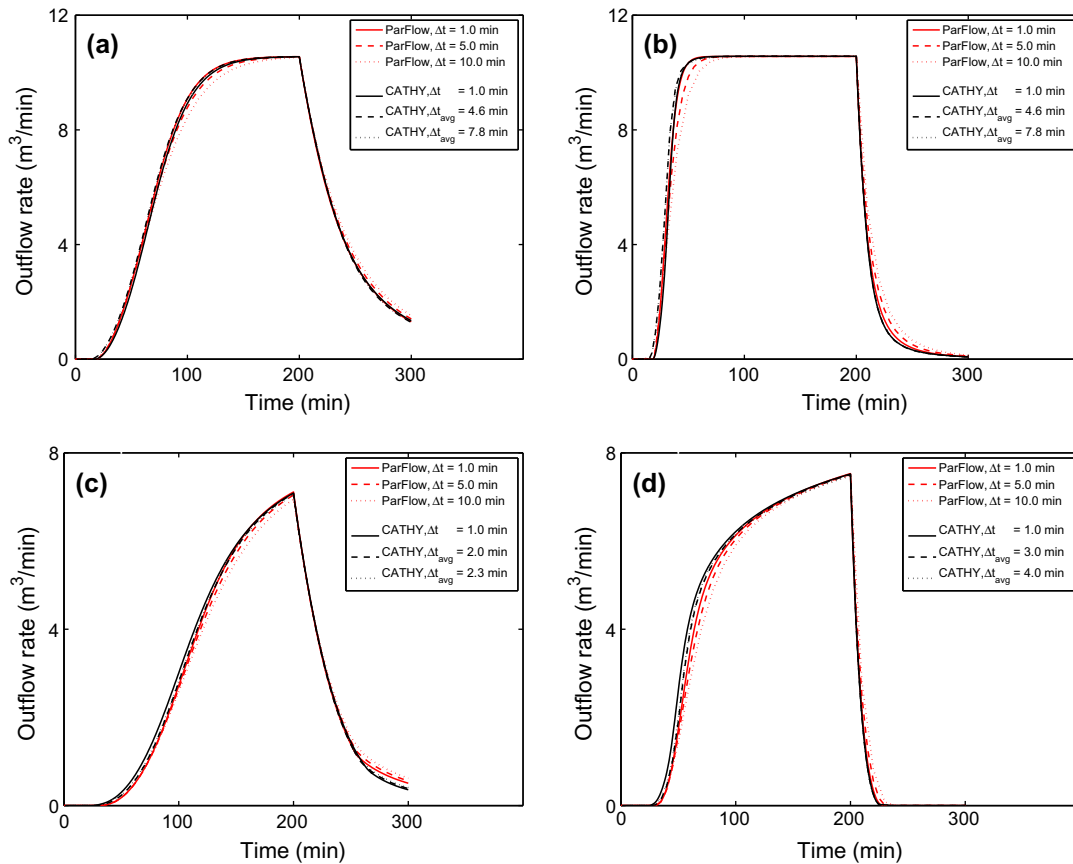


Fig. 4. Influence of slope angle and time step size for the homogeneous sloping plane test case. Top: saturation excess runoff generation (with initial water table depth of 0.5 m) for slope angles 0.05% (a) and 5% (b). Bottom: infiltration excess runoff generation (with $K_s = 6.94 \times 10^{-5}$ m/min) for slope angles 0.05% (c) and 5% (d).

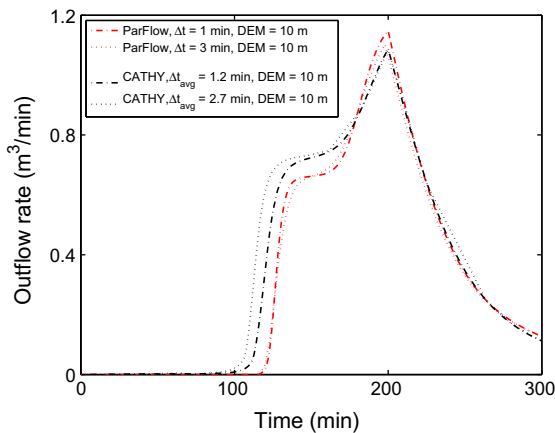


Fig. 5. Outflow response for the heterogeneous sloping plane test case at 10 m DEM resolution under variable (CATHY) and uniform (ParFlow) time discretizations.

in the mesh strip immediately upslope of the slab and by the higher value reached at the first rising limb of the hydrograph. This effect is due to the loss of accuracy in passing information from nodes (subsurface) to cells (surface) and can be alleviated by a finer horizontal discretization, as can be seen in Fig. 6. The higher value reached by the CATHY model at the first rising limb of the hydrograph is again connected to the different spatial discretization schemes implemented in the two models. This results in a smaller volume of infiltrated water above the slab for CATHY, causing in turn a lower degree of saturation beneath the slab.

The differences in behavior of the ParFlow and CATHY models at and in the vicinity of the slab can be seen more clearly in Fig. 7, where, for the coarse DEM case, vertical profiles of saturation (S_w) are shown 60 min into the simulation for the five points indicated in Fig. 6. The profiles are identical or very similar at the downslope edge of the slab ($x = 250$ m, high wetness due to re-infiltration from the slab) and farther away from the slab in the upslope direction ($x = 140$ m, dry profiles). The ParFlow and CATHY profiles are most different in the center of the slab ($x = 200$ m, saturated at the surface and a sharp drop in wetness below the surface), at the upslope edge of the slab ($x = 150$ m), and further downslope of the slab ($x = 280$ m), due to greater re-infiltration front spreading for the CATHY model.

4.1.3. Return flow dynamics, homogeneous conditions

In this set of simulations the dynamics of return flow processes is evaluated by tracking the evolution of the point of intersection (or exit point) between the water table and the land surface during a storm-interstorm simulation consisting of 200 min of rain followed by 200 min of evaporation. The high conductivity values used for the test case (6.94×10^{-2} m/min) allow a rapid rise and fall of the water table and a consequently fast expansion and contraction of the sloping plane's partial contributing area. Two slope angles of 0.5% and 5.0% were simulated, with an initial water table depth of 0.5 m and a constant time step of 0.5 min. For each of the slope angles, in addition to the ParFlow simulations (representing a sheet flow parameterization of overland routing), two CATHY simulations were run, with surface parameterizations corresponding to sheet flow and rill flow (these parameterizations are described in the next subsection).

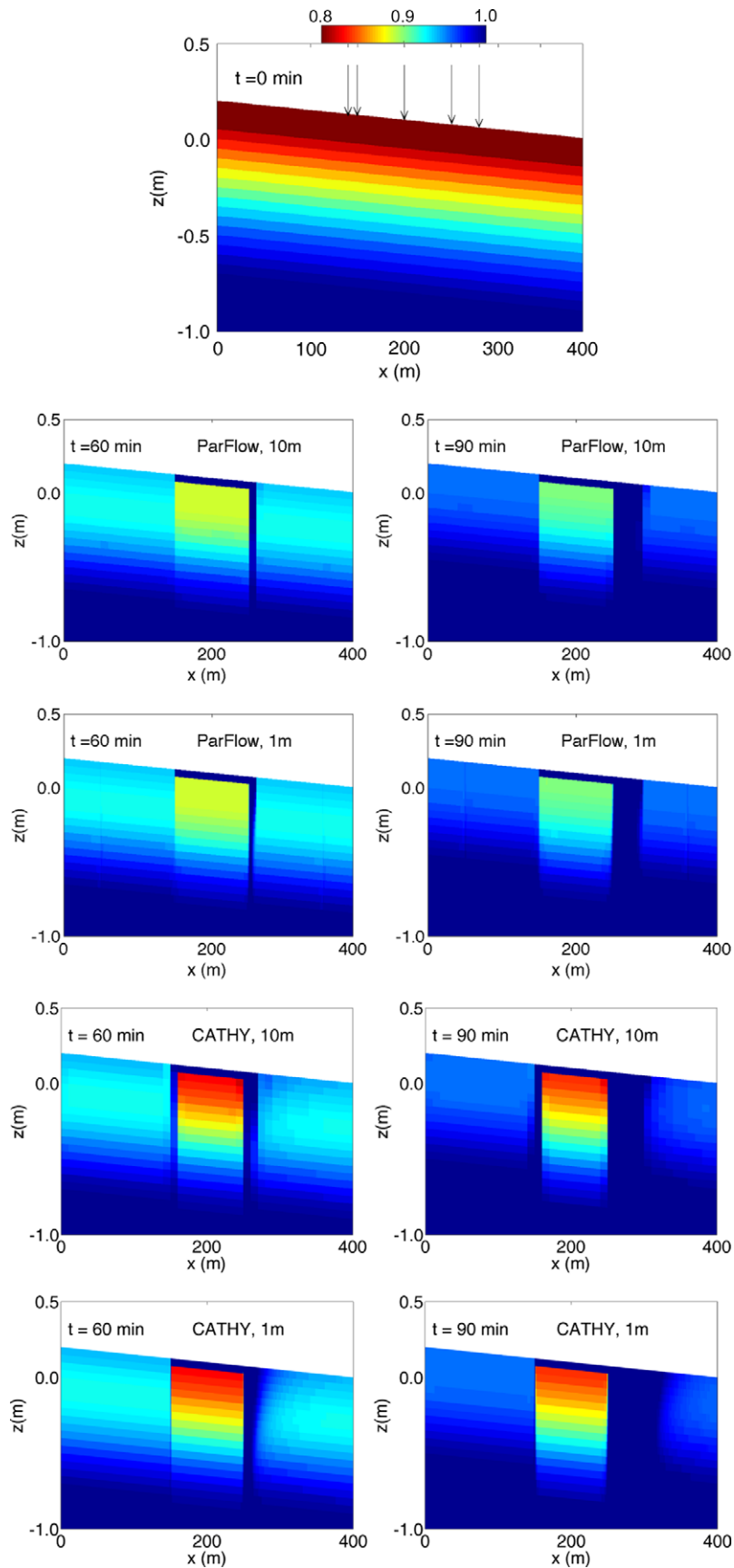


Fig. 6. Saturation (S_w) profiles in cross section (at $y = 200$ m) for the heterogeneous sloping plane test case at grid cell resolutions of 10 m and 1 m and with a variable time discretization. In the top figure the initial conditions are shown, as well as the location along the transect of the five vertical profiles plotted in Fig. 7.

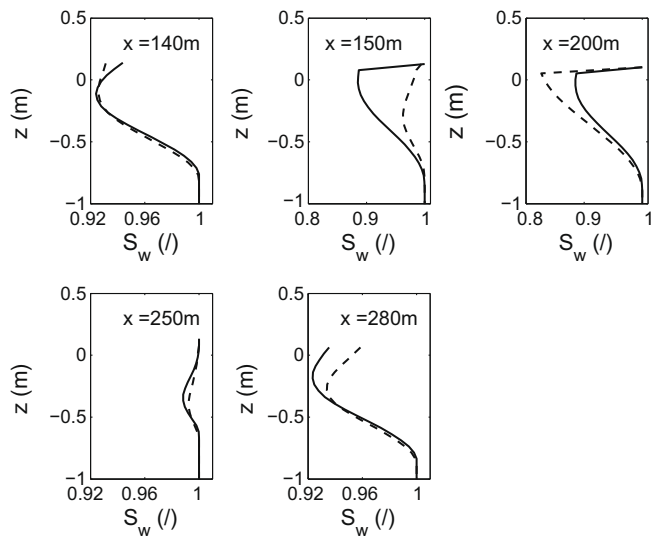


Fig. 7. Vertical saturation profiles for the heterogeneous sloping plane test case at the five locations indicated in Fig. 6. The output is taken at time 60 min, the simulations used a grid cell resolution of 10 m, and the temporal discretization was variable. ParFlow results are the solid lines and CATHY results are the dashed lines.

The simulation results are shown in Fig. 8 for the transect in the middle of the sloping plane. At gentler slopes, infiltration and subsurface flow remain predominantly vertical, thus the water table rise is relatively uniform over the catchment, and slower than at the downslope nodes of the steeper case. It thus takes longer for the exit point to move upslope, but when it does so it is more abrupt than for the steeper slope, and it reaches a higher position upslope (saturating almost the entire catchment). When rainfall ceases, on the other hand, gentler-sloping catchments drain more slowly than steep catchments, and thus we observe that during the recession phase the exit point recedes downslope more slowly than for the steep plane. As can be seen in Fig. 8, ParFlow and CATHY agree quite well for both slopes in the timing of the water table intersection with the ground surface and in its evolution during the entire rainfall period. A more marked discrepancy (especially for the 0.5% slope) was instead found for the evaporation period, with the CATHY model sheet flow case characterized by a slower downward recession of the exit point. This discrepancy may be attributed to the different parameterizations of the routing scheme diffusion process in the two models. The close match be-

tween ParFlow and both CATHY parameterizations during the rainfall period clearly indicates that the dynamics of the return flow process is controlled by the subsurface domain during this phase, whereas surface routing dominates the process during the recession phase. It is interesting to observe that, despite the differences in ParFlow and CATHY response during the recession period, the ParFlow result nonetheless lies within the envelope defined by the CATHY rill and sheet flow extremes.

4.1.4. Overland flow routing

The representation of surface runoff as two-dimensional sheet flow or as one-dimensional rill flow can have a strong effect on ponding heads, surface flow velocities, and outlet hydrographs [13], and on connected overland and channel flow processes such as soil erosion and sediment transport [27,14,15]. In this next set of simulations on the sloping plane (with DEM resolution 80 m and slope angle 0.05%), the influence of hydraulic geometry parameters was investigated, as well as the possibility of setting “equivalent” parameterizations between the sheet flow conceptualization used in ParFlow and the rill flow representation in CATHY.

In CATHY the water-surface width of each cell of the river network is scaled in space as a function of the upstream drainage area according to the following relationship:

$$W(A, 1) = W(A_s, Q_f) Q_f (A_s)^{-b'} (A/A_s)^{w(b''-b')}, \quad (15)$$

where $W(A, 1)$ is the water-surface width at a site draining area A for a unit-flow discharge. This unit-flow water-surface width is dynamically varied in time as a function of the simulated flow discharge Q by multiplying Eq. (15) by $Q^{b'}$ (see Eq. (7)). In the simulations presented here we set $W(A_s, Q_f)$ to 80 m, with $A_s = 3.2 \times 10^4 \text{ m}^2$ being the drainage area at the outlet cell and $Q_f = 10.56 \text{ m}^3/\text{min}$ being the steady state outlet discharge obtained from a simulation with b' and b'' equal to zero, i.e., with a water-surface width of 80 m over the entire catchment and through the entire simulation. This set of values, with the water-surface width constant and equal to the DEM resolution, should correspond most closely to a sheet flow representation. With these values of A_s , Q_f , and $W(A_s, Q_f)$, different surface width parameterizations were then simulated in CATHY by using different values of the Leopold and Maddock Jr. [22] “at-a-station” (b') and “downstream” (b'') scaling exponents.

The results of these tests are presented in Fig. 9, on the left for fixed b'' (0.5) and varying b' (from 0.0 to 0.5 in increments of 0.1) and on the right for fixed b' (0.0) and varying b'' (from 0.0 to 0.5 in increments of 0.1). When $b' = 0.0$ the water-surface width is time-invariant and when $b' = b''$ it is space-invariant. The upper

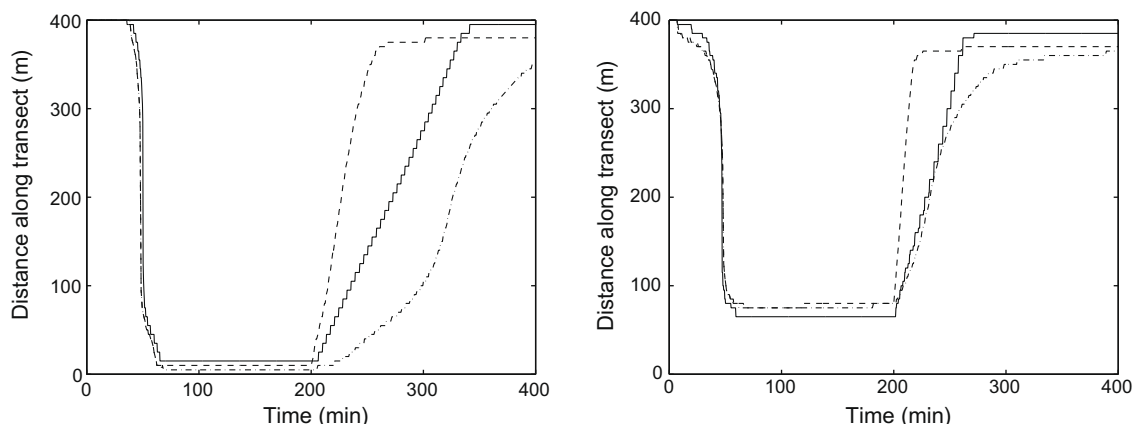


Fig. 8. Evolution of the point of intersection between the water table and the land surface for the sloping plane test case with slope angles 0.5% (left) and 5.0% (right). The outlet face is at $x = 400 \text{ m}$. ParFlow results are the solid lines and CATHY results are the dashed-dotted lines (sheet flow) and dashed lines (rill flow).

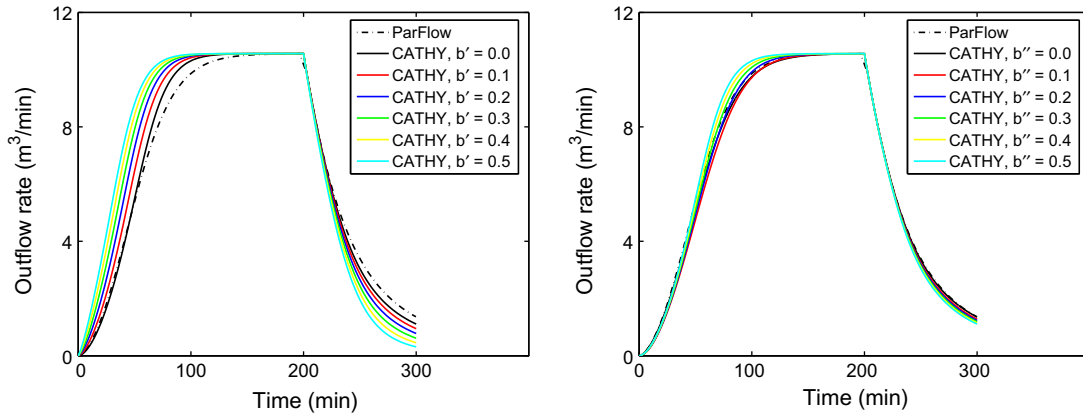


Fig. 9. Sloping plane test case: comparison of outflow responses for sheet flow (ParFlow) and rill flow (CATHY) surface routing models. For CATHY a range of “at-a-station” (b' , left) and “downstream” (b'' , right) Leopold and Maddock Jr. [22] scaling exponents was used.

value of 0.5 for both the b' and b'' ranges is taken from Orlandini and Rosso [32]. For increasing b' we get decreasing water-surface widths, and thus faster routing and hydrograph response, as can be seen in Fig. 9. For increasing b'' (with $b' = 0$) we also get decreasing water-surface widths and again faster hydrograph response. Fig. 9 shows that the CATHY model is more sensitive to the “at-a-station” scaling exponent than to the “downstream” exponent, which is to be expected for a sloping plane, where the drainage area varies linearly with elevation.

As alluded to above, the combination $b' = 0$, $b'' = 0$ in Fig. 9 does indeed produce the closest match with the ParFlow (sheet flow) results. It is thus possible in this test case to parameterize a rill flow conceptualization that accurately mimics a sheet flow model. It is notable that this good match was obtained despite the different approximations of the Saint-Venant equation (kinematic and diffusion waves) implemented in the ParFlow and CATHY models, and despite the gentle slope of the plane catchment, which favors diffusion-dominated overland flow (compensated for in kinematic wave models by the introduction of numerical diffusion [4]).

4.2. Tilted V-catchment

For the tilted V-catchment rill and sheet flow conceptualizations are again examined. In this case three different water-surface width parameterizations for the CATHY model are compared against the sheet flow results obtained with ParFlow. In the first parameterization a sheet flow representation was used for the entire catchment, in the second a rill flow representation, and in the third case sheet flow was used for the channel cells and rill flow for the hillslope cells. In the first case we set b' and b'' to zero and $W(A_s, Q_f)$ to 20 m (equal to the DEM resolution), with $A_s = 1.62 \times 10^6 \text{ m}^2$ being the drainage area of the entire catchment and $Q_f = 291.6 \text{ m}^3/\text{min}$ being the steady state outlet discharge obtained for this combination of parameter values. In the rill flow parameterization for the entire catchment (second case) we set $b' = 0.36$ and $b'' = 0.50$, with, for the hillslope portion of the catchment surface, $W(A_s, Q_f) = 6.5 \text{ m}$ (water-surface width at the uppermost channel cell at steady state), $A_s = 1.6 \times 10^4 \text{ m}^2$ (the drainage area at the hillslope outlet cell), and $Q_f = 2.88 \text{ m}^3/\text{min}$ (the steady state discharge at the hillslope outlet cell), and for the channel portion of the catchment domain, $W(A_s, Q_f) = 20 \text{ m}$, $A_s = 1.62 \times 10^6 \text{ m}^2$, and $Q_f = 291.6 \text{ m}^3/\text{min}$. In the third case (combined sheet and rill flow) we set $b' = b'' = 0$ for the channel cells and $b' = 0.36$, $b'' = 0.50$ for the hillslope cells, with $W(A_s, Q_f) = 20 \text{ m}$ (water-surface width at channel cells at steady state condition), $A_s = 1.6 \times 10^4 \text{ m}^2$, and

$Q_f = 2.88 \text{ m}^3/\text{min}$ for the hillslope cells and $W(A_s, Q_f) = 20 \text{ m}$, $A_s = 1.62 \times 10^6 \text{ m}^2$, and $Q_f = 291.6 \text{ m}^3/\text{min}$ for the channel cells.

The results in Fig. 10 for the ParFlow model and for the three surface routing parameterizations in CATHY show that the ParFlow hydrograph lies between the CATHY rill flow case (fastest response) and the CATHY sheet flow case (slowest response). The sheet flow and combined sheet flow/rill flow responses are very similar, and are closer to ParFlow's sheet flow hydrograph, suggesting that the channel flow segment of the catchment dominates the response for this test case. The differences between rill flow and sheet flow dynamics are highlighted in Fig. 11, which traces the evolution of ponding heads (surface flow depths) along the channel during the first 30-min of rainfall for CATHY (three different parameterizations) and for ParFlow. The flow depths are lowest for the sheet flow conceptualization, about 50% higher for the combined sheet/rill flow case, and about two times higher (relative to the combined case) for the rill flow parameterization. The energy gradients are also much stronger in the rill flow case, in particular along the first 600 m of the channel reach. This is due to the non-linearity of the stage-discharge relationship for the three simulated flow conditions. These differences in water depth profiles (steeper in space and faster in time for the rill flow parameterization) underscore the differences in erosion potential and sediment transport capacity between the two conceptualizations.

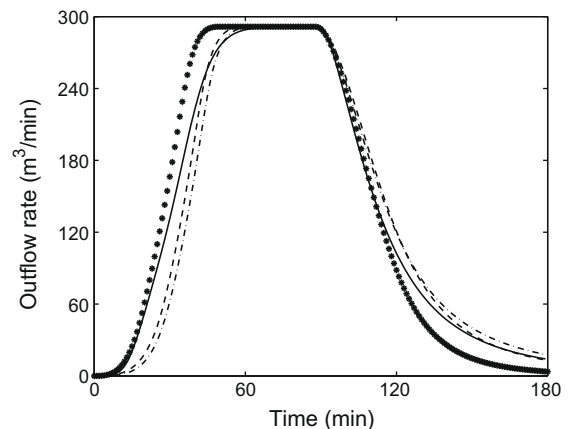


Fig. 10. Outflow response for the tilted V-catchment test case: ParFlow (solid line), CATHY with sheet flow parameterization (dashed-dotted line), CATHY with rill flow parameterization (dotted line), and CATHY with combined sheet and rill flow parameterization, for the channel and hillslope, respectively (dashed line).

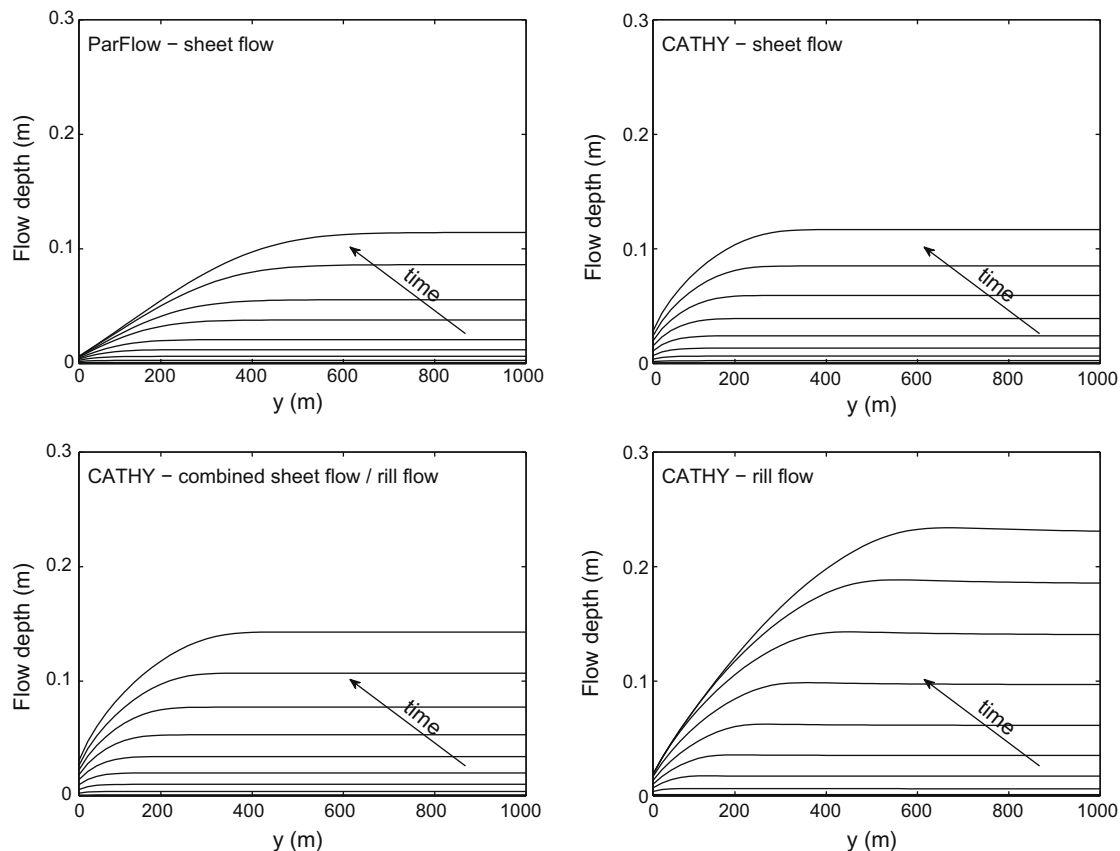


Fig. 11. Tilted V-catchment test case: evolution of surface flow depth along the channel ($x = 800$ m) during the first 30 min of the rainfall period for CATHY (using three different surface routing parameterizations) and for ParFlow. The catchment outlet is at $y = 1000$ m.

5. Conclusions

Two physically-based, spatially-distributed models of conjunctive surface and subsurface flow, ParFlow and CATHY, have been compared. The analysis has been focused on examining the coupling approaches implemented in the two models and the different conceptualizations used to describe the propagation of surface runoff. The theoretical and numerical bases for ParFlow and CATHY were briefly presented, highlighting key features and differences between the models, and two test problems were simulated under a variety of configurations and parameter combinations to explore runoff generation, surface routing, and other dynamics, and the impact of heterogeneity, grid resolution, and other factors on this dynamics.

ParFlow and CATHY were both found to be quite sensitive to vertical discretization for scenarios of infiltration excess runoff generation, with some differences in behavior between the two models due also to the different discretization schemes used (finite difference and finite element). For both models, convergence of the solutions under mesh refinement was demonstrated. The CATHY model was found to be more sensitive to temporal discretization, in particular for the heterogeneous test case, owing to the sequential coupling approach used in this model. It was shown that under “equivalent” sheet flow parameterizations the CATHY and ParFlow models produce very similar hydrograph responses at the catchment outlet, and differences in CATHY model response under rill flow and sheet flow conceptualizations were highlighted.

A broader goal of this study is to encourage additional efforts to assess coupled surface–subsurface models, given the variety of formulations and approaches that have recently been proposed for representing flow interactions and other processes. Suitable

benchmark test cases will need to be devised to extend the work presented here, for instance larger catchments with more complex geometry and topography.

Acknowledgments

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